MIL-F-8209B
2 Feb 1971
SUPERSEDING
MIL-F-8209A(ASG)
2 August 1961

MILITARY SPECIFICATION

FLASHLIGHT, HAND GENERATED, TYPE A-9
(LAMP ASSEMBLY, FLASHLIGHT, TYPE A-9)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of flashlight, designated type A-9.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Set-Up Paperboard

Military

MIL-P-116	Preservation, Methods Of
MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 First article. This specification makes provisions for first article inspection (see 4.4).

3.2 Component parts. The flashlight shall consist of a plastic or metal housing containing a generator, reflector, and lamp socket. It shall have an external lever for operating the generator.

3.3 Material.

* 3.3.1 Specifications and standards. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraphs.

3.3.1.1 Standard parts. AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins, may be used, provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.3.2 Protective treatment. When materials are used in the construction of the flashlight that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction. The flashlight shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 The flashlight shall be constructed so that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases.

3.5 Performance. The flashlight shall conform to this specification when subjected to the applicable tests of Section 4.

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* 3.6 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

3.7 Details of components.

3.7.1 The complete flashlight shall be similar to the one shown on Figure 1. However, the sample shown is a representative sample, and assemblies of a slightly different design shall be considered as conforming to the requirements, provided the flashlight is in accordance with the dimensions and all other requirements specified herein.

* 3.7.2 Lamp head. The lamp head shall be designed to use a 2.7 volt (V), 0.15 ampere, B-3-1/2, clear bulb, miniature, flange-base, U.S.A.S.I. No. PR-9 lamp. The lamp shall be mounted so that it shall be held firmly in position. The service lamp and one spare lamp shall be furnished by the manufacturer with each flashlight. Means shall be provided in each lamphead housing for storing the spare lamp.

* 3.7.2.1 Light beam. The lamp head, as shown on Figure 1, shall be provided with a parabolic reflector suitably plated and polished, and when used with a lamp as specified herein and operated at 2.7V with the lens in place, shall produce a light beam equal to or in excess of that shown on Figure 2. The reflector shall be securely held in the lens ring but shall be easily removed by hand.

3.7.2.2 The lamp head with reflector shall be provided with an unbreakable lens of clear plastic.

3.7.2.3 The design of the lamp head shall permit relamping without the use of tools.

3.7.3 Housing and gears. The lamp gear housing, lens retainer, and the case housing shall be fabricated from injection molded type cellulose base plastic or suitable metal.

* 3.7.3.1 The color shall be olive drab conforming to Color No. 14087 of FED-STD-595.

3.7.3.2 The frame of the generator shall be sufficiently strong to hold the bearings in alignment.

3.7.3.3 The generator and gear train bearings shall be made of self-oiling material.

3.7.4 Power unit. The power unit shall consist of an alternating-current generator with a stationary armature and a 4 or more pole rotating permanent magnet field of a highly retentive material. The generator shall be driven by a hand-operated lever with a reciprocating movement which shall actuate a rack or gear segment.

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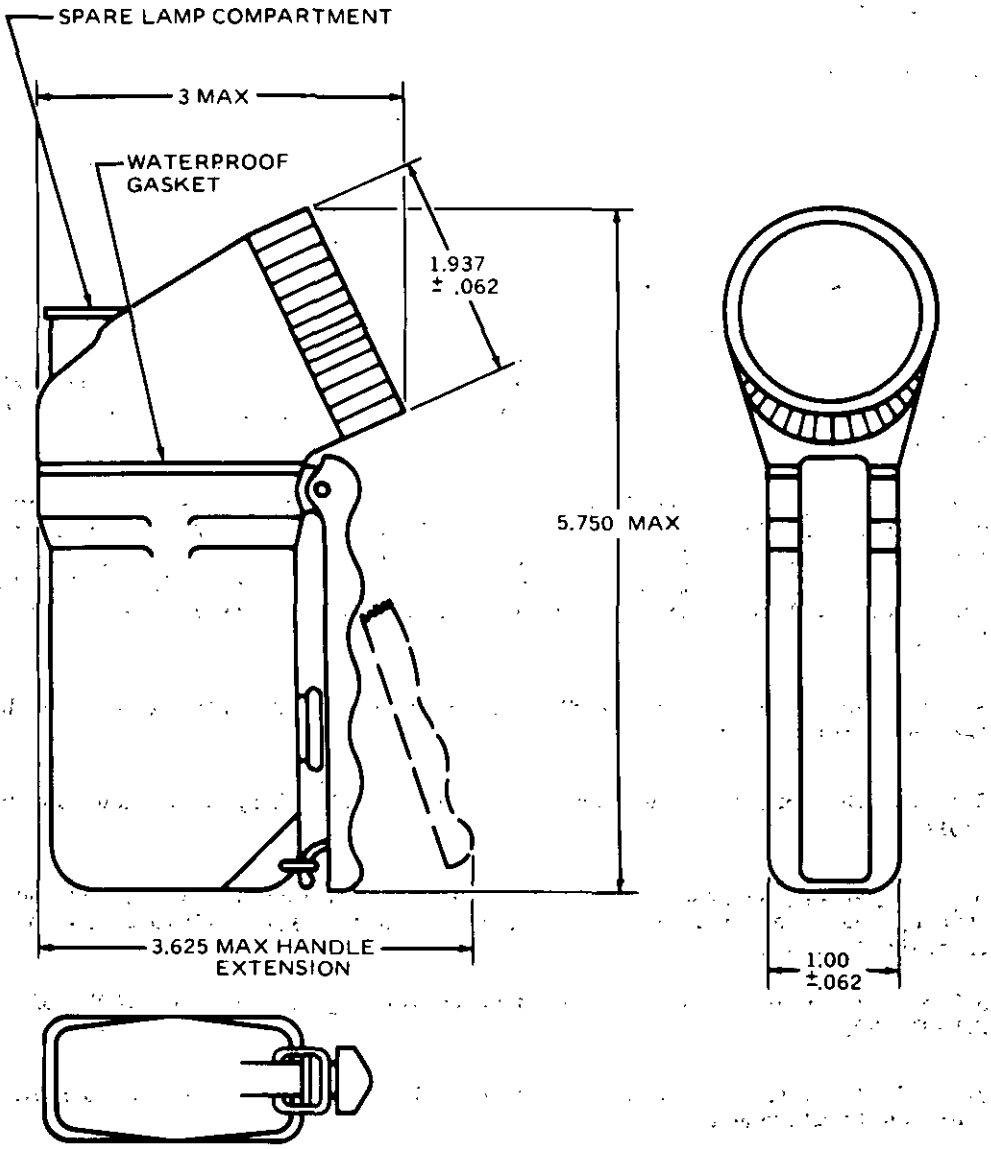


FIGURE 1. FLASH LIGHT ASSEMBLY

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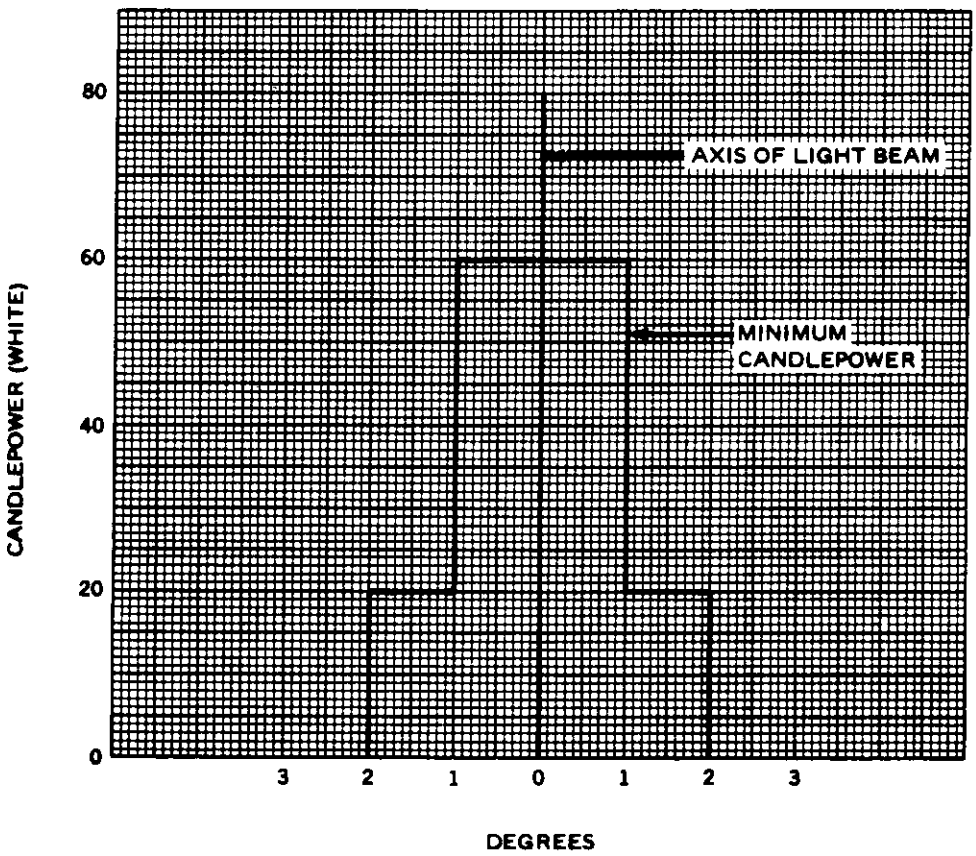


FIGURE 2. LIGHT BEAM INTENSITY

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3.7.5 Operating lever. The operating lever shall have the fulcrum located toward the lamp head end of the flashlight, and a means shall be provided for locking the handle in a closed position when the flashlight is not in use.

3.7.6 Operating voltage. The flashlight shall be capable of maintaining a root mean square (rms) voltage output of not less than 2.1V and not more than 3.5V when the lever is operated 120 even strokes per minute and the generator is loaded with a shunt resistor and voltmeter in a combined resistance of 14.2 ohms, (the approximate resistance of the specified lamp at rated voltage). At no speed of operation that may reasonably be attained shall the voltage exceed 3.5V when the generator is loaded in the above manner. The term "even strokes" is intended to mean that approximately one-half of the stroke cycle is used in closing the lever at a uniform rate and approximately one-half is used in allowing the lever to reopen.

3.7.7 Operating force. The operating force measured at a point 3 inches from the fulcrum of the operating lever shall not exceed 10 pounds when the lever is actuated at a rate of 60 to 120 even strokes per minute.

3.8 Dimensions. The dimensions shall be as shown on Figure 1.

3.9 Weight. The weight of the flashlight shall not exceed 0.80 pound.

3.10 Finishes and protective coatings.

3.10.1 All exposed metal parts shall be made of corrosion-resistant material or suitably plated to prevent corrosion.

* 3.10.2 Aluminum-alloy parts. Aluminum-alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows.

3.10.2.1 Small holes and case inserts need not be anodized.

* 3.10.2.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

* 3.10.2.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatment in accordance with MIL-C-5541 may be used in lieu of anodizing.

* 3.10.2.4 Castings containing nonaluminum-alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541.

* 3.10.2.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

* 3.11 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.12 Workmanship. The flashlight, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, and wiring; alignment of parts and tightness of assembly screws, and bolts.

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3.12.1 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.12.2 Cleaning. The flashlight shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specifed in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of the flashlight shall be classified as follows:

a. First article inspection (4.4)

b. Quality conformance inspection (4.5)

4.3 Test conditions.

* 4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not definitely specified, the test shall be made at atmospheric pressure (approximately 29.92 inches mercury) and at room temperature (approximately 25° Centigrade (C)). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

* 4.4 First article inspection.

* 4.4.1 First article samples. The first article samples shall consist of three complete flashlights representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity (see 6.2).

* 4.4.2 Tests. The first article inspection shall consist of all the tests specified in this specification. Tests shall be conducted in the following order:

Sample No. 1	Sample No. 2	Sample No. 3
Individual tests	Individual tests	Individual tests
Photometric	High temperature	Vibration
Operating force	Low temperature	
Operating voltage	Humidity	
	Sand and dust	
	Salt spray	

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4.4.2.1 Photometric. The sample flashlight shall be subjected to a photometric test to determine compliance with the light distribution requirements of 3.7.2.1. The lamp shall be operated at rated voltage for this test. Failure to conform to these requirements shall be cause for rejection.

4.4.2.2 Operating force. The flashlight shall be tested to determine whether the applied force at the lever is within the limits specified in 3.7.7.

4.4.2.3 Operating voltage. The flashlight shall be operated and the rms voltage measured with the generator loaded as specified in 3.7.6. Failure of the flashlight to maintain the voltages specified in 3.7.6, when operated at 120 even strokes per minute, shall be cause for rejection.

4.4.2.4 Environmental tests.

4.4.2.4.1 Low temperature. The flashlight shall be subjected to a temperature of -54°C for 4 hours and tested. The test shall consist of operating the flashlight while at -54°C, and after it has returned to room temperature. Failure of the flashlight to operate at full brightness at -54°C shall not be considered as failure to operate.

4.4.2.4.2 High temperature. The flashlight shall be subjected to a temperature of 71°C dry heat for a period of 24 hours. After being subjected to this temperature, the flashlight shall be compared with an untested sample for dimensional stability, crazing, and other defects. Failure of the flashlight to be in full working condition during and after this test shall be cause for rejection.

* 4.4.2.4.3 Salt spray. The flashlight shall be subjected to a salt spray test in accordance with Procedure I of MIL-E-5272. The flashlight shall be removed from the test chamber and operated for 50 strokes at approximately 24-hour intervals. The test shall consist of 3 such 24-hour periods. The flashlight shall operate satisfactorily at the conclusion of the 72-hour exposure, and no part shall show evidence of corrosion which is detrimental to its operation. Failure in this test shall be cause for rejection.

4.4.2.4.4 Humidity. The flashlight shall be tested for resistance to humidity in accordance with Procedure III of MIL-E-5272. Improper operation of the assembly at the conclusion of this test shall be cause for rejection.

* 4.4.2.4.5 Sand and dust. The flashlight shall be tested for resistance to sand and dust in accordance with Procedure I of MIL-E-5272, except that the test shall consist only of one test period of 6 hours at 25°C. The assembly shall operate at the conclusion of this test and failure to do so shall be cause for rejection.

* 4.4.2.4.6 Vibration. The flashlight shall be tested in accordance with Procedure XII of MIL-E-5272. The flashlight shall not be operated during the vibration test; however, it shall be subjected to the operation test (see 4.5.1.2) after completion of the vibration test. Any failure shall be cause for rejection.

* 4.5 Quality conformance inspection. Quality conformance inspection shall consist of individual tests.

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4.5.1 Individual tests. Each flashlight shall be subjected to the following tests.

4.5.1.1 Examination of product. The flashlight shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.5.1.2 Operation. The flashlight shall be tested for proper operation.

* 4.6 Packaging, packing, and marking. Preparation for delivery shall be examined for conformance with 5.1 thru 5.3.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

* 5.1.1 Level A. Each flashlight shall be adequately cushioned and individually packaged Method III of MIL-P-116. The unit container shall conform to PPP-B-676.

* 5.1.2 Level C. Flashlights shall be packaged in a manner which affords adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

* 5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.2).

* 5.2.1 Level A. Flashlights packaged as specified by the contract or order shall be packed in fiberboard shipping containers conforming to PPP-B-636, class weather resistant. Insofar as practicable, shipping containers shall contain identical quantities, be similar in shape and size, and, of maximum cube consistent with the protection required. Container closure, sealing, and strapping shall be in accordance with the appendix to PPP-B-636.

* 5.2.2 Level B. Level B shall be the same as Level A, except shipping containers shall conform to class domestic.

* 5.2.3 Level C. Flashlights shall be packed in a manner which affords adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

* 5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The type A-9 flashlight covered by this specification is intended for use on aircraft emergency liferafts.

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* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

c. Where the first article samples should be sent and instructions concerning the submittal of test reports (see 4.4.1).

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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